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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM101 - AUTOMOTIVE SYSTEMS FOR E - VEHICLES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 marks

1. (d) increase
2. (b)To convert AC voltage into DC voltage
3. (c)Energy losses during transmission
4. (b)A simple amplifier without feedback
5. (c)By switching the current in the stator windings using an electronic controller
6. b). Signal amplification
7. a). The output voltage becomes unstable
8. b). It reduces the power to maintain the speed
9. a). The LED corresponding to that threshold voltage turns on
10. c). 80% SOC

Part B
Short Answer Questions
Answer any 4 questions
Each question carries 5 marks

11. An operational amplifier (op-amp) is a high-gain, DC-coupled electronic voltage amplifier that uses external feedback components (resistors/capacitors) to perform mathematical operations, signal conditioning, and filtering. It amplifies the difference between its two inputs,inverting and non-inverting

Basic Functionality of an Op-Amp

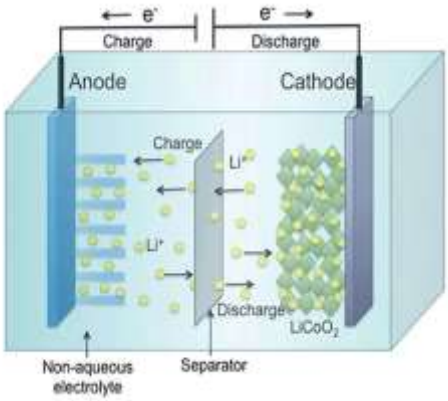
- Differential Input
- Voltage Amplification
- Feedback Mechanism
- Three-Terminal Operation

Ideal Characteristics of an Op-Amp

- Infinite Open-Loop Gain
- Infinite Input Impedance
- Zero Output Impedance
- Infinite Bandwidth
- Infinite Common-Mode Rejection Ratio

- Zero Input Offset Voltage
- Zero Propagation Delay

12. Working with diagram can give 5 marks



Lithium-ion batteries operate by moving lithium ions between a cathode and anode through an electrolyte, typically using a constant-current/constant-voltage (CC/CV) charge profile and a reverse flow during discharge. During charging, ions move from cathode to anode; during discharge, they move back, creating a flow of electrons that powers devices

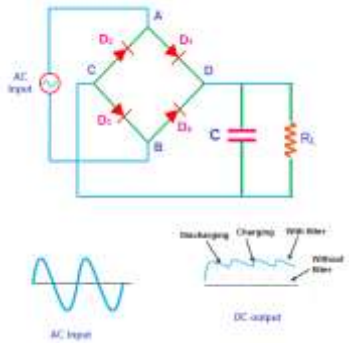
Charging Process (Storing Energy)

- **Ion Movement:** When a charger is connected, it forces lithium ions to leave the positive cathode (typically Li-metal oxide) and travel through the electrolyte and separator to the negative graphite anode.
- **Electron Flow:** Electrons move through the external circuit from the charger to the anode, balancing the positive charge of the incoming ions.

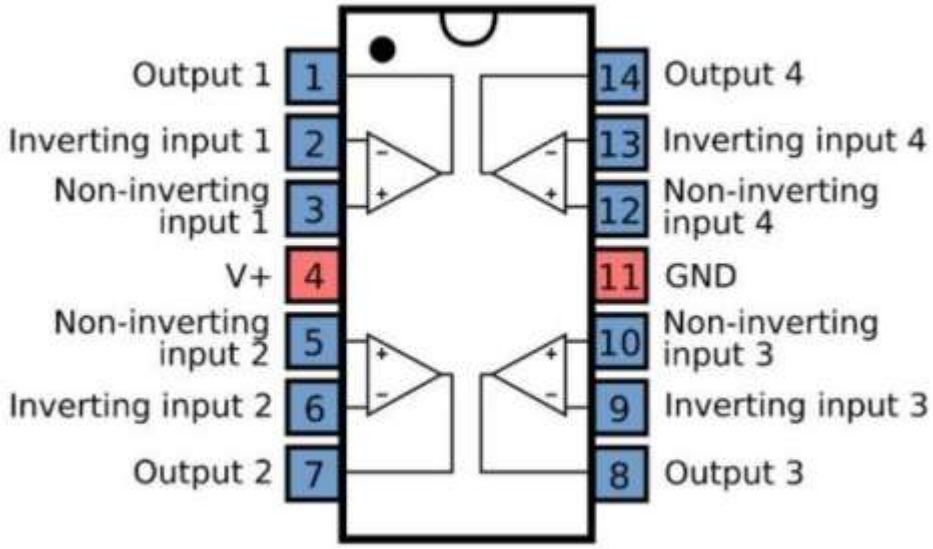
Discharging Process (Releasing Energy)

- **Ion Movement:** When a device is connected, the chemical energy is released, and lithium ions move from the anode back to the cathode through the electrolyte.
- **Electron Flow:** Electrons move from the anode to the cathode through the external circuit, powering the connected load.

13. A capacitor filter works in a bridge rectifier by smoothing the pulsating DC output into a steadier, smoother DC voltage. Connected in parallel with the load resistor, the capacitor charges to the peak voltage during the diode conduction periods and discharges through the load when the rectifier output drops, effectively filling in the gaps between pulses.

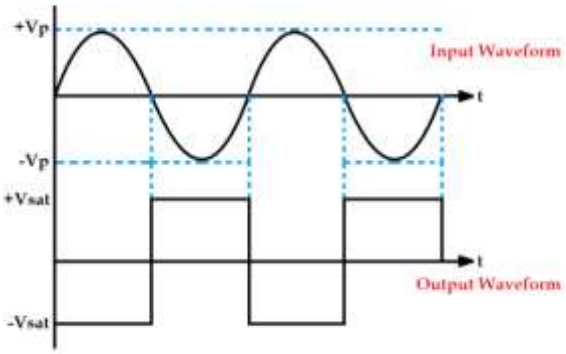


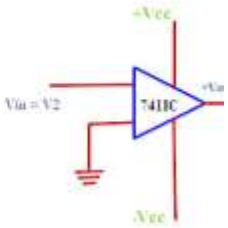
14. Pin Configuration LM324



15. Comparator as zero crossing detector

To use a comparator as a **Zero Crossing Detector (ZCD)**, you essentially set the reference voltage to zero by connecting one of its inputs directly to the ground. The comparator then switches its output state whenever the input signal crosses this zero-volt threshold, effectively converting a sine wave into a square wave.





16.

- **Trickle Charge**
- **Constant Current (CC)**
- **Constant Voltage (CV)**
- **Overcharge Protection**
- **Thermal Regulation**
- **Under-Voltage Lockout (UVLO)**
- **Over-Discharge Protection**.
- **Short-Circuit & Over-Current Protection**

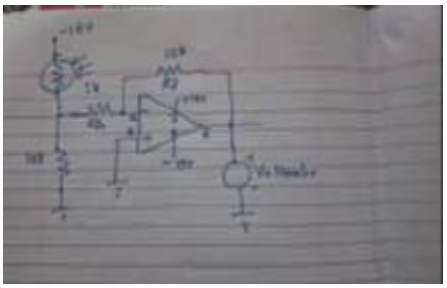
Part C

Essay Questions

Answer any 2 questions

Each question carries 10 marks

17. **LDR (Light Dependent Resistor), also known as a photoresistor, is a passive component whose resistance changes based on the intensity of light hitting its surface. In the dark, its resistance is very high (often several megohms), but when exposed to bright light, the resistance drops significantly**

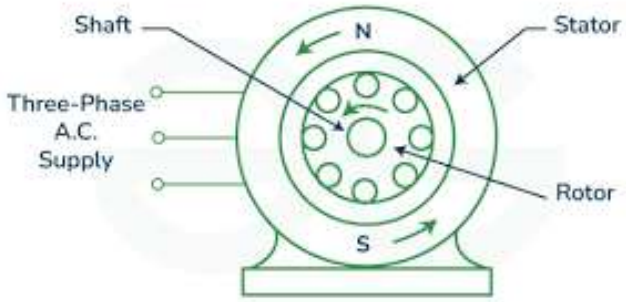


18. **An induction motor operates on the principle of electromagnetic induction, converting electrical energy into mechanical energy without any direct electrical connection to the rotating part.**

    Stator: The stationary part producing the rotating magnetic field.

    Rotor: The rotating component, either squirrel-cage or wound, that receives induced current.

    An induction motor works on the principle of Faraday's Law of Electromagnetic Induction, where a stator creates a rotating magnetic field (RMF) when fed by a three-phase AC supply. This RMF cuts the rotor conductors, inducing an EMF and current, which interacts with the RMF to produce torque and turn the rotor.



19. Input and output capacitors are essential for ensuring the stability and performance of a 78XX voltage regulator, primarily by preventing oscillations and managing rapid load changes

    Input Capacitor

    Stability & Filter: The input capacitor is primarily used to ensure the regulator remains stable and does not self-oscillate. [

- **Preventing Oscillation:** If the regulator is far from the power source (typically more than 10 inches), the lead inductance can cause the IC to oscillate. A small capacitor (typically 0.33 µF) placed close to the input pin bypasses this inductance.
- **Filtering Ripple**: It acts as a reservoir to smooth out fluctuations or AC ripples from the incoming DC supply, ensuring the regulator receives a "clean" signal to process.
  Output Capacitor:
  Transient Response: The output capacitor acts as a local energy buffer for the connected load.
- **Improved Transient Response:** The 78XX is a relatively "slow" regulator from the 1970s and cannot respond instantly to sub-microsecond changes in load current. If a load suddenly demands more current, provides that energy immediately, preventing the output voltage from sagging.
- **Noise Reduction:** It filters out high-frequency noise that may be generated internally by the regulator or externally by the load, resulting in a cleaner DC output

20. A **unity gain amplifier**, also commonly known as a **voltage follower** or **buffer amplifier**, is an operational amplifier (op-amp) circuit where the output voltage is exactly equal to the input voltage
Gain of a Unity Gain Amplifier
The voltage gain of a unity gain amplifier is **1 (unity)**. This means there is no amplification or attenuation of the signal amplitude; the signal simply "follows" the input

